NE763: ADVANCED REACTOR MATERIALS

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EXAM 2: SFRs

DYNTI DAS

1 DESCRIBE SOME OF THE BENEFICIAL SAFETY FEATURES OF SFRs

- INHERENT STABILITY → AS $T_{coolant}$ ↑, CORE EXPANDS WHICH RESULTS IN INCREASED NEUTRON LEAKAGE, REDUCES THE CHAIN REACTION, THEN RESULTS IN $T_{coolant}$ ↓ AND THE CORE SHUTSDOWN W/ NO HUMAN INTERVENTION
- DUE TO LARGE SODIUM POOL, SFRs ALSO HAVE A SELF-COOLING CAPABILITY TO REMOVE DECAY HEAT W/ NO HUMAN INTERVENTION
- HIGHER THERMAL CONDUCTIVITY THAN WATER
- CAN BE OPERATED AT ATMOSPHERIC PRESSURES DUE TO LARGE OPERATING RANGE
  ↳ LOW MELTING POINT ~ 371K, HIGH BOILING POINT ~ 1154K

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2 WHY DO WE ALLOY URANIUM?

- TO REDUCE FUEL SWELLING AND SUBSEQUENT DAMAGE TO CLADDING/CORE COMPONENTS.
- PURE U FUEL PRODUCE FISSION PRODUCT GASES → α-U FUEL IS AMORPHORICALLY? AND ANISOTROPIC WITH INTERSTITIAL CRYSTALINE STRUCTURES ON DIFFERENT PLANES → AS FP GASES + PORES MIGRATE, THEY MIGRATE DIFFERENTLY ON DIFFERENT PLANES → RESULTS IN STRESSES WHICH ARE ASSUMED BY THE GRAIN BOUNDARY AND RESULTS IN CAVITATION (OCCURS BETWEEN 400-600 K)
- ALLOYING URANIUM FUEL REDUCES CAVITATION AND HELPS MAINTAIN CORE/FUEL INTEGRITY + LONGEVITY
  - stabilize γ, ↑ melting point, fcc, etc.

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3 WHY MUST METALLIC FUEL SYSTEMS INCLUDE A RELATIVELY LOW SMEAR DENSITY AND LARGE PLENUM?

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- TO PROVIDE A VOLUME SPACE FOR FUEL EXPANSION + SWELLING DUE FP BUILDUP. ~75% SMEAR DENSITY IN METALLIC FUELS ALLOWS FOR FUEL SWELLING WHICH REDUCES THE STRESSES PLACED ON THE CLADDING. ADDITIONALLY, THIS LARGE PLENUM PROVIDES A SPACE FOR BUILD-UP OF FP GASES TO VENT TO THE RELIEVE THE PRESSURE BUILD UP IN THE FUEL SLUG. THIS PROMOTES INCREASED CLADDING LIFE. ALSO SUPPORTS THERMAL CONDUCTIVITY IN THE FUEL → INITIAL THERMAL CONDUCTIVITY ↓ AS FP GASES ↑ IN THE FUEL → AS GASES GET VENTED TO PLENUM, THAT SPACE IS FILLED BY Na FROM BOND SODIUM WHICH ↑ THERMAL CONDUCTIVITY.

- why is 75% enough? bubbles interconnect & release

4 WHAT IS CONSTITUENT REDISTRIBUTION IN METALLIC FUELS? WHY DOES IT OCCUR? WHAT ARE ITS IMPACTS?

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- IT IS THE MIGRATION OR MOVEMENT OF Zr OR U IN THE FUEL RESULTING IN VARYING CONCENTRATIONS ACROSS THE FUEL.
- THE DRIVING FACTOR IS TEMPERATURE GRADIENTS ACROSS THE FUEL → Zr MOVES TOWARD HOTTER REGIONS WHILE U MOVES TOWARDS COLDER REGIONS
- IMPACTS ARE CHANGES IN THERMAL CONDUCTIVITY ACROSS THE FUEL, FUEL DISTRIBUTION, POROSITY CHANGES, AND CHEMICAL INTERACTION IMPACTS.
- IMPACTS FUEL CLADDING MECHANICAL INTERACTIONS → URANIUM MOVING TOWARD CLADDING BOUNDARIES ↑ FCMIs
- FUEL CLADDING CHEMICAL INTERACTIONS ALSO IMPACTED

- also chemical potential, different Zr solubility in different phases

5 HOW DOES THERMAL CONDUCTIVITY VARY AS A FUNCTION OF BURNUP IN METALLIC FUELS? WHAT PHENOMENA DRIVE THIS BEHAVIOR?

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- INITIAL THERMAL CONDUCTIVITY WILL ↓ AS FUEL BURNUP COMMENCES ↑ FP GASES FORM WITHIN THE FUEL. INITIALLY FP GASES TAKE UP THE ADDITIONAL SPACE IN THE FUEL AS DESIGNED BY THE ~75% SMEAR DENSITY. WHEN FP GASES BUILDUP REACHES A CERTAIN PRESSURE IT WILL VENT TO THE GAS PLENUM IN THE FUEL PIN, SEPARATELY, SODIUM FROM THE BOND SODIUM WILL DIFFUSE INTO THE FUEL AND REPLACE THE SMEAR VOIDS LEFT BY THE FP GASES + DEPLETED FUEL. SINCE Na HAS A DECENT THERMAL CONDUCTIVITY, OVERALL FUEL CONDUCTIVITY WILL RAPIDLY ↑, COUNTERING THE INITIAL ↓ IN CONDUCTIVITY.

6 WHAT IS FCCI? WHAT ARE THE ADVERSE EFFECTS OF FCCI? WHAT ARE THE PRIMARY FUEL AND CLADDING SPECIES PARTICIPATING IN FCCI?

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- FCCI IS FUEL CLADDING CHEMICAL INTERACTIONS THAT OCCUR AT THE FUEL/CLADDING INTERFACE RESULTING CHEMICAL COMPOSITION CHANGES AND PRODUCTION OF COMPOUNDS.
- ADVERSE EFFECTS IS THAT IT CAN RESULT IN EMBRITTLING THE CLADDING, INCREASING CORROSION, DEVELOPING LOCALIZED HOT SPOTS WHICH RESULT IN MALDISTRIBUTION of FLUX AND HEAT PRODUCTION AND PRODUCES UNWANTED TEMPERATURE GRADIENTS ACROSS THE FUEL AND CLADDING.
- PRIMARY SPECIES PARTICIPATING ARE Zr, U, OXYGEN, Ni, Cr, Pu.
  - Fe, Cr from cladding, Ce, Nd from fuel
  - low melting phases
  - needed more...

7 DESCRIBE RESTRUCTURING OF MOX FUELS, INCLUDING WHY IT HAPPENS

- RESTRUCTURING OF MOX FUELS OCCURS DUE TO OXIDES MIGRATING W/ THE FUEL TO THE FUEL BOUNDARY. MOX FUELS START HYPOSTOICHIOMETRIC, 1.97~2.00, AND AS FUEL BURNUP OCCURS, OXIDES MIGRATE TO THE PERIPHERY MAKING THE PERIPHERY MORE STOICHIOMETRIC AND THE CENTER LESS OXIDIZED. THE TEMPERATURE GRADIENT IS WHAT DRIVES THIS MIGRATION.
- THIS CAN CAUSE INCREASED CORROSION IN THE PERIPHERY.
- ALSO PORES/VOIDS (MIGRATE) FOUR DIFFERENT REGIONS → COMBINED, GRANULAR, EQUIAXED, AND AS SINTERED. AS PORES TRAVEL, THEY CONDENSELY FORM IN THE COMBINED REGION, TAKE A MORE ELONGATED PATH IN THE GRANULATED (COLUMNAR) REGION, CREATE A MORE POROUS STRUCTURE IN THE EQUIAXED REGION AND POROSITY REMAINS SIMILAR TO INITIAL DESIGN IN THE AS SINTERED REGION. PORES TRAVEL DUE CONDENSATION ON THE COOLER SIDE AND EVAPORATION ON THE HOTTER SIDE OF A GAS BUBBLES/PORE → DRIVES PORES TO THE CENTER OF THE FUEL

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- this is not the term I used, so I needed you to explain
- got pieces of it, but not a clear picture here

8 HOW DOES Pu AND O VARY SPATIALLY IN A MOX PIN?

- Pu AGGLOMORATES DUE TO IMPURITIES WHILE MACHINING THE FUEL THEREFORE THIS RESULTS LOCALIZED AREAS W/ HIGH Pu CONCENTRATION.
- DUE TO THE HYPOSTOICHIOMETRIC INITIAL DESIGN → Pu IS CONCENTRATED TOWARD THE CENTER AND SPREAD AXIALLY.
- FOR OXYGEN, IT IS SPREAD THROUGHOUT TO MAINTAIN M/O RATIO AND SPREADS RADIALLY → ↑ at r↑

6/10

- pore migration

9 WHAT IS JOG? WHY DOES IT OCCUR? WHAT ARE ITS IMPACTS?

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- JOINT OXYDE GAIN.
- OCCURS WHEN INSOLUBLE METAL OXIDES FORM FROM FISSION PRODUCTS AND THEY FORM ON CLADDING SURFACES DUE TO THE INCREASED OXYGEN POTENTIAL.
- IMPACTS ARE OVERALL POSITIVE → PROVIDES ABILITY TO ABSORB FUEL SWELLING, PREVENT CORROSION FOR A PERIOD OF TIME, IMPROVES THERMAL CONDUCTIVITY. mechanical buffer
- NEGATIVE IMPACTS → TOO MUCH CAN RESULT IN LOCALIZED HOT SPOTS FOR CORROSION AND CAUSE CLADDING EMBRITTLEMENT.
  - leads to FCCI

10 DISCUSS THE MEANS IN WHICH SODIUM CORROSION OCCURS IN SFRS.

8/8

- TYPE I CORROSION: SOLUBLE COMPOUNDS DISSOLVE IN HOTTER REGIONS BUT THEN PRECIPITATE IN COOLER REGIONS → FOR SFRS, WEIGHT REDUCTION DUE TO Na CORROSION OCCURS IN FUEL REGION AND THEN WEIGHT INCREASE OCCURS IN HX REGION
- TYPE II CORROSION: MORE CHEMICALLY DRIVEN CORROSION DUE TO SOLUBLE METALS, IMPURITIES, TEMPERATURE GRADIENTS, DECARBURIZATION/CARBURIZATIONS, SODIUM FLOW VELOCITY.